

Sets, Functions, and Cardinality

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COMP2711 Discrete Mathematical Tools

Overview

Suppose every student in a class needs a locker.

- **Sets** specify the students and lockers.
- A **function** assigns each student to one locker.
- Special functions tell us whether lockers are shared or unused.

Functions will later let us compare the sizes of infinite sets.

1 Sets

1.1 Membership and set descriptions

A **set** is an unordered collection of distinct objects. The objects are its **elements**. We write

$$x \in A \quad \text{when } x \text{ is an element of } A, \quad x \notin A \quad \text{otherwise.}$$

Throughout this course,

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}, \quad \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\},$$

and $\mathbb{R}$ is the set of real numbers: all points on the number line.

There are two common ways to describe a set.

Method	Description	Example
List all elements	write every element	$O = \{1, 3, 5, 7, 9\}$
Set builder	state a membership test	$O = \{n \in \mathbb{N} \mid n < 10 \text{ and } n \text{ is odd}\}$

Set-builder notation needs a domain. The expression

$$\{x \in D \mid P(x)\}$$

contains exactly the elements x in D for which $P(x)$ is true. It is the **truth set** of P over D . Restricted quantifiers use the same domain convention:

$$\forall x \in D P(x) \equiv \forall x (x \in D \rightarrow P(x)), \quad \exists x \in D P(x) \equiv \exists x (x \in D \wedge P(x)).$$

Sets may contain strings, numbers, or even other sets. For example,

$$\{\text{"Ada"}, \text{"Ben"}\}, \quad \{-2, 0, \tfrac{1}{2}, \sqrt{2}\}, \quad \{7, \text{"hello"}, \{1, 2\}\}.$$

1.2 Set equality

MUST UNDERSTAND A standard set is unordered and deduplicated.

Two sets are equal when they have exactly the same elements:

$$A = B \iff \forall x (x \in A \leftrightarrow x \in B).$$

Order and repetition do not matter:

$$\{1, 3, 5\} = \{5, 3, 1, 3\}.$$

1.3 Membership and subsets

The symbols $\in$ and $\subseteq$ have different roles. We say that B is a **subset** of A when every element of B also belongs to A :

$$B \subseteq A \iff \forall x (x \in B \rightarrow x \in A).$$

Now the equality test can also be written

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

A **proper subset** is a subset that is not equal to the full set:

$$B \subsetneq A \iff B \subseteq A \text{ and } B \neq A.$$

Thus, A contains at least one element not in B .

For $A = \{1, \{2, 3\}\}$,

$$1 \in A, \quad \{1\} \subseteq A, \quad \{2, 3\} \in A, \quad 2 \notin A.$$

The braces matter: 1 is a number, while $\{1\}$ is a set.

1.4 Set size

A finite set's **size**, also called its **cardinality**, is the number of distinct elements. It is written $|S|$. For example,

$$|\{a, b, c\}| = 3, \quad |\{1, 1, 2\}| = |\{1, 2\}| = 2.$$

The **empty set** is $\emptyset = \{\}$. It is the only set of size zero:

$$|\emptyset| = 0.$$

1.5 Power sets

The **power set** $\mathcal{P}(S)$ is the set of all subsets of S :

$$\mathcal{P}(S) = \{T \mid T \subseteq S\}.$$

For $S = \{a, b\}$:

Choice bits	00	01	10	11
Subset	$\emptyset$	$\{b\}$	$\{a\}$	$\{a, b\}$

The first bit decides whether a is included; the second decides whether b is included.

If $|S| = n$, then

$$|\mathcal{P}(S)| = 2^n.$$

Why? A subset makes one independent yes/no choice for each element of S : include it or exclude it. Thus every subset corresponds to one bit string of length n .

1.6 Cartesian products

The **Cartesian product** of A and B is

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

For $A = \{1, 2\}$ and $B = \{x, y, z\}$,

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}.$$

If A and B are finite, then $|A \times B| = |A||B|$. More generally,

$$A_1 \times \cdots \times A_n = \{(a_1, \dots, a_n) \mid a_i \in A_i \text{ for every } i\}.$$

Each element is an **ordered pair**:

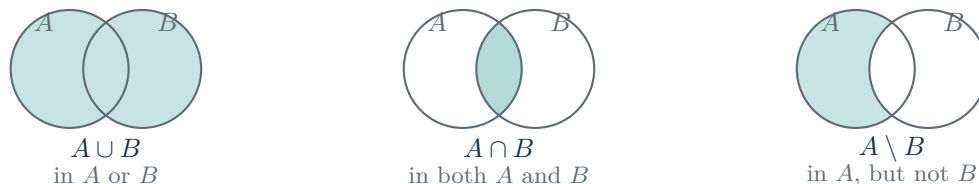
$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

2 Set operations

A **universal set** U contains all objects under discussion. The four basic operations are:

Operation	Set	Membership condition
Union	$A \cup B$	$x \in A$ or $x \in B$ (or both)
Intersection	$A \cap B$	$x \in A$ and $x \in B$
Difference	$A \setminus B$	$x \in A$ and $x \notin B$
Complement	$\bar{A} = U \setminus A$	$x \notin A$

A **Venn diagram** draws sets as regions. The shaded region shows the result.



Sets A and B are **disjoint** when $A \cap B = \emptyset$. For finite sets, the overlap must be subtracted once when a union is counted:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

2.1 Set identities

The membership conditions give a direct translation:

$$\cap \leftrightarrow \wedge, \quad \cup \leftrightarrow \vee, \quad x \in \overline{A} \leftrightarrow \neg(x \in A).$$

Therefore propositional equivalences produce set identities. For example,

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

is De Morgan's law.

Worked proof: an identity element by element

Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Let x be arbitrary. Then

$$\begin{aligned} x \in A \setminus (B \cup C) &\iff x \in A \wedge x \notin B \cup C \\ &\iff x \in A \wedge \neg(x \in B \vee x \in C) \\ &\iff x \in A \wedge x \notin B \wedge x \notin C \\ &\iff (x \in A \setminus B) \wedge (x \in A \setminus C) \\ &\iff x \in (A \setminus B) \cap (A \setminus C). \end{aligned}$$

Because this holds for every x , the two sets are equal.

Other useful identities include

$$\begin{aligned} A \cup \emptyset &= A, & A \cap U &= A, \\ A \cup \overline{A} &= U, & A \cap \overline{A} &= \emptyset, \\ A \cup (B \cap C) &= (A \cup B) \cap (A \cup C), & A \cap (B \cup C) &= (A \cap B) \cup (A \cap C). \end{aligned}$$

Rather than memorizing an isolated table, translate membership to logic when an identity is uncertain.

2.2 Families of sets

For a family $\{A_i\}_{i \in I}$,

$$\bigcup_{i \in I} A_i = \{x \mid \exists i \in I (x \in A_i)\}, \quad \bigcap_{i \in I} A_i = \{x \mid \forall i \in I (x \in A_i)\}.$$

Read the first as “the elements appearing in at least one set” and the second as “the elements appearing in every set.”

2.3 Computer representation

For $U = \{u_1, \dots, u_n\}$, a subset of U can be stored as n bits. Bit i is 1 exactly when u_i is in the subset. Then union, intersection, and complement become bitwise OR, AND, and NOT.

A **multiset** also records repetitions. Ordinary sets do not:

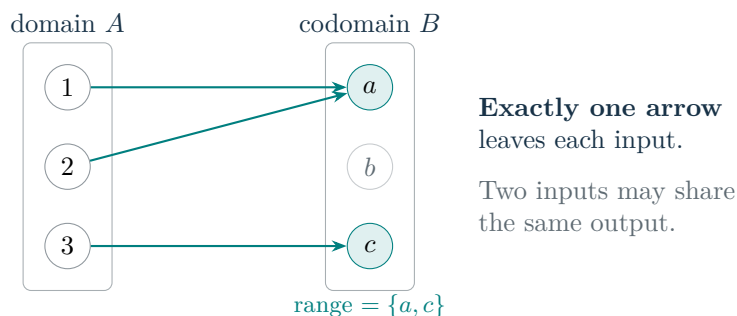
$$\{a, a, b\} = \{a, b\} \quad \text{as sets.}$$

3 Functions

3.1 Definition of a function

MUST UNDERSTAND Every input has exactly one output.

A **function** $f : A \rightarrow B$ assigns exactly one element of B to each element of A . The set A is the **domain**; B is the **codomain**.



For $f : A \rightarrow B$:

Domain	A
Codomain	B
Range	elements in B touched by f : $f[A]$

$$f[A] = \{f(a) \mid a \in A\} \subseteq B.$$

For example, let

$$f : \{1, 2, 3\} \rightarrow \{a, b, c\}, \quad f(1) = a, \quad f(2) = a, \quad f(3) = c.$$

Its range is $\{a, c\}$. The element b is in the codomain but not the range.

For $S \subseteq A$, its **image** is

$$f[S] = \{f(s) \mid s \in S\}.$$

For $T \subseteq B$, its **preimage** is

$$f^{-1}[T] = \{a \in A \mid f(a) \in T\}.$$

This preimage exists for every function; it is not an inverse function.

The **graph** of f records all input–output pairs:

$$G_f = \{(a, f(a)) \mid a \in A\} \subseteq A \times B.$$

An assignment is not a function if an input has no output or more than one output. Different inputs may share an output.

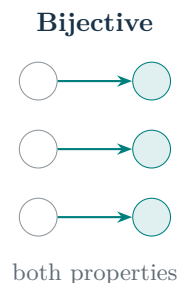
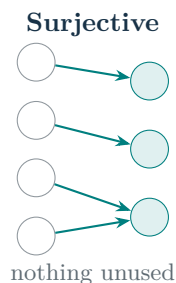
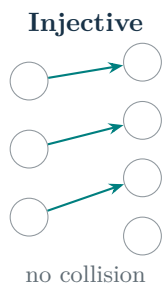
A **total function** is defined on every element of its domain. A **partial function** may be undefined for some inputs. In this course, “function” means total function unless stated otherwise.

3.2 Injection, surjection, and bijection

MUST UNDERSTAND Separate collisions from unused targets.

Let $f : A \rightarrow B$.

Property	Mathematical definition	Plain meaning
Injective	$\forall a_1, a_2 \in A, f(a_1) = f(a_2) \rightarrow a_1 = a_2$	no collisions
Surjective	$\forall b \in B, \exists a \in A, f(a) = b$	nothing unused
Bijjective	injective and surjective	exactly one input per target



The injective definition has an equivalent contrapositive form:

$$\forall a_1, a_2 \in A, \quad a_1 \neq a_2 \rightarrow f(a_1) \neq f(a_2).$$

Equal outputs force equal inputs; equivalently, distinct inputs have distinct outputs.

Squaring on the integers

Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be $f(n) = n^2$.

It is not injective because $f(1) = f(-1)$. It is not surjective because, for example, -1 has no integer preimage. Its range is

$$f[\mathbb{Z}] = \{0, 1, 4, 9, \dots\}.$$

Now let

$$g : \mathbb{N} \rightarrow \{0, 1, 4, 9, \dots\}, \quad g(n) = n^2.$$

If $g(m) = g(n)$, then $m^2 = n^2$. Since $m, n \geq 0$, we get $m = n$. Every target is $k^2 = g(k)$ for some $k \in \mathbb{N}$. Thus g is bijective.

The algebraic rule did not change. The domain and codomain did.

3.3 Inverse functions

A function $f : A \rightarrow B$ has an **inverse function** $f^{-1} : B \rightarrow A$ exactly when f is bijective. It reverses every match:

$$f^{-1}(b) = a \iff f(a) = b.$$

For example, let $h : \mathbb{Z} \rightarrow \mathbb{Z}$ be $h(x) = x + 1$. Then $h^{-1}(y) = y - 1$.



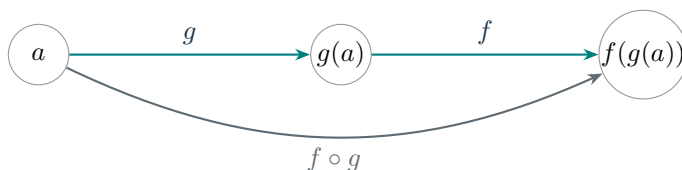
The number positions stay fixed. The inverse reverses the arrows.

3.4 Composition

If $g : A \rightarrow B$ and $f : B \rightarrow C$, their **composition** is

$$f \circ g : A \rightarrow C, \quad (f \circ g)(a) = f(g(a)).$$

Read $f \circ g$ from right to left: apply g first, then f .



For $g(x) = x + 1$ and $f(x) = 2x$ on $\mathbb{Z}$,

$$(f \circ g)(x) = 2(x + 1) = 2x + 2.$$

Composition laws

Composition is **associative** whenever the compositions are defined:

$$(f \circ g) \circ h = f \circ (g \circ h).$$

The grouping changes, but the order remains h , then g , then f .

Composition is generally **not commutative**. For the functions above,

$$(f \circ g)(x) = 2x + 2, \quad (g \circ f)(x) = 2x + 1.$$

Therefore $f \circ g \neq g \circ f$.

3.5 Floor and ceiling

For $x \in \mathbb{R}$, the **floor** and **ceiling** are

$$\lfloor x \rfloor = \text{the greatest integer at most } x, \quad \lceil x \rceil = \text{the least integer at least } x.$$

For example,

$$\lfloor 2.7 \rfloor = 2, \quad \lceil 2.7 \rceil = 3, \quad \lfloor -2.7 \rfloor = -3, \quad \lceil -2.7 \rceil = -2.$$

Equivalently,

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1, \quad \lceil x \rceil - 1 < x \leq \lceil x \rceil.$$

3.6 Counting functions

Let $|A| = n$ and $|B| = m$.

For an unrestricted function $A \rightarrow B$, each of the n inputs independently chooses one of m outputs. Therefore there are

$$m^n$$

functions.

For an injective function, outputs cannot be reused. When $m \geq n$, there are

$$m(m-1) \cdots (m-n+1) = \frac{m!}{(m-n)!}$$

injections. When $m < n$, there are none. In particular, two n -element sets have $n!$ bijections between them.

3.7 Practice

Set practice

Let $A = \{1, 2\}$ and $B = \{2, 3\}$. Compute $A \cup B$, $A \cap B$, $A \setminus B$, $\mathcal{P}(A)$, and $A \times B$.

Solution.

$$\begin{aligned} A \cup B &= \{1, 2, 3\}, & A \cap B &= \{2\}, & A \setminus B &= \{1\}, \\ \mathcal{P}(A) &= \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}, & A \times B &= \{(1, 2), (1, 3), (2, 2), (2, 3)\}. \end{aligned}$$

Function practice

Let $f, g : \mathbb{Z} \rightarrow \mathbb{Z}$ be $f(n) = n + 3$ and $g(n) = 2n$.

1. Find $(f \circ g)(n)$ and $(g \circ f)(n)$.
2. Decide whether each function is injective, surjective, or bijective.
3. Find the inverse of any bijective function.

Solution.

$$(f \circ g)(n) = 2n + 3, \quad (g \circ f)(n) = 2n + 6.$$

The function f is bijective, with $f^{-1}(n) = n - 3$. The function g is injective but not surjective: it reaches the even integers and misses the odd integers.

4 Cardinality: Describing the size of sets

The **cardinality** of a set describes its size. For a finite set, this is usually easy to understand: count its elements. For example,

$$|\{a, b, c, d\}| = 4.$$

Infinite sets are different. We cannot finish counting their elements one by one, and a proper subset may have the same size as the whole set. Instead, we compare sizes by asking whether the elements of two sets can be matched.

4.1 Comparing cardinalities

MUST UNDERSTAND Compare cardinalities by constructing injections, surjections, and bijections.

For sets A and B , functions give three direct comparisons:

Function from A to B	Cardinality	Meaning
injection	$ A \leq B $	no two elements of A collide
surjection	$ A \geq B $	every element of B is reached
bijection	$ A = B $	no collisions and every element of B is reached

The **Schroeder–Bernstein theorem** says that injections in both directions guarantee equal cardinality:

$$|A| \leq |B| \text{ and } |B| \leq |A| \implies |A| = |B|.$$

Equivalently, if there is an injection $f : A \rightarrow B$ and an injection $g : B \rightarrow A$, then there is a bijection between A and B . The proof is non-trivial and is outside the scope of this lecture.

Example 1: Natural numbers and even numbers

Let

$$\text{Even} = \{0, 2, 4, 6, \dots\} \subsetneq \mathbb{N}.$$

The inclusion map $i : \text{Even} \rightarrow \mathbb{N}$, given by $i(x) = x$, is injective. Therefore $|\text{Even}| \leq |\mathbb{N}|$.

There is also an injection in the other direction:

$$g : \mathbb{N} \rightarrow \text{Even}, \quad g(x) = 2x.$$

It is injective because

$$x_1 \neq x_2 \implies 2x_1 \neq 2x_2,$$

and it is surjective onto Even because every even natural number has the form $2x$. Hence g is bijective and

$$\boxed{|\mathbb{N}| = |\text{Even}|.}$$

A proper subset of an infinite set can have the same cardinality as the full set.

Example 2: An interval and the real line

We prove $|(0, 1)| = |\mathbb{R}|$ in two steps. First,

$$g : (0, 1) \rightarrow (0, \infty), \quad g(x) = \frac{1}{x} - 1$$

is a bijection with inverse $g^{-1}(y) = 1/(y + 1)$. Second,

$$f : (0, \infty) \rightarrow \mathbb{R}, \quad f(y) = \ln y$$

is a bijection with inverse $f^{-1}(z) = e^z$.

The composition is therefore a bijection:

$$h = f \circ g : (0, 1) \rightarrow \mathbb{R}, \quad h(x) = \ln\left(\frac{1}{x} - 1\right).$$

Its inverse is $h^{-1}(z) = 1/(1 + e^z)$. Thus

$$\boxed{|(0, 1)| = |\mathbb{R}|.}$$

4.2 Countable sets

MUST UNDERSTAND A countable set can be listed so that every element appears.

Intuitive meaning. A set S is **countable** if we can list all its elements as

$$s_1, s_2, s_3, \dots$$

and every element of S appears somewhere in the list. A finite list may stop.

Formal definition. A set S is countable if it is finite, or if there is a bijection

$$e : \mathbb{N} \rightarrow S$$

that lists all its elements. Equivalently,

$$\boxed{|S| \leq |\mathbb{N}|.}$$

Countable case 1: The integers

Theorem. The integers $\mathbb{Z}$ are countable.

To prove $|\mathbb{Z}| \leq |\mathbb{N}|$, it is enough to construct a surjection $e : \mathbb{N} \rightarrow \mathbb{Z}$. The following function is in fact bijective. Define $e : \mathbb{N} \rightarrow \mathbb{Z}$ by

$$e(0) = 0, \quad e(2k-1) = k, \quad e(2k) = -k \quad (k \geq 1).$$

This gives the list

$$0, 1, -1, 2, -2, 3, -3, \dots$$

Every integer occurs: 0 occurs first, positive k occurs at index $2k-1$, and negative $-k$ occurs at index $2k$. These index formulas are distinct, so no integer repeats. Hence e is bijective and $\mathbb{Z}$ is countably infinite.

Countable case 2: The rational numbers

Theorem. The rational numbers $\mathbb{Q}$ are countable.

The rational numbers are

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

To prove $|\mathbb{Q}| \leq |\mathbb{N}|$, it is enough to find a surjection $e : \mathbb{N} \rightarrow \mathbb{Q}$: a list in which every rational number appears.

Every positive rational number has a representation p/q with positive integers p and q . Arrange the pairs (p, q) by increasing $p+q$:

$$(1, 1); \quad (1, 2), (2, 1); \quad (1, 3), (2, 2), (3, 1); \quad \dots$$

Traverse each finite diagonal and list the value p/q only when it has not appeared earlier. Every positive rational appears on some diagonal, and duplicates such as $1/1 = 2/2$ are skipped. This produces a list $q_1, q_2, \dots$ of all positive rationals, with no omissions and no repetitions.

Now alternate each positive rational with its negative:

$$0, q_1, -q_1, q_2, -q_2, \dots$$

This list contains every rational number exactly once, so $\mathbb{Q}$ is countable.

Countable case 3: Cartesian products

Theorem. If A and B are countable, then $A \times B$ is countable.

Suppose the countably infinite sets A and B have lists

$$A = \{a_0, a_1, a_2, \dots\}, \quad B = \{b_0, b_1, b_2, \dots\}.$$

Every pair in $A \times B$ has the form (a_i, b_j) . List the pairs by increasing $i + j$:

$$(a_0, b_0); \quad (a_0, b_1), (a_1, b_0); \quad (a_0, b_2), (a_1, b_1), (a_2, b_0); \quad \dots$$

Each diagonal is finite, and every pair occurs on exactly one diagonal. Therefore $A \times B$ is countable. The same conclusion holds if either factor is finite. In particular,

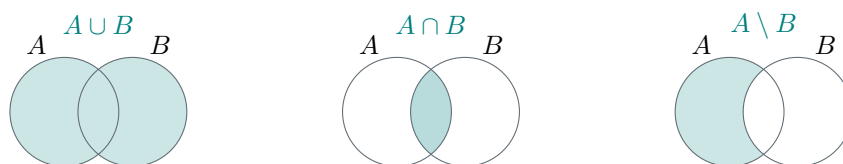
$\mathbb{Q} \times \mathbb{Q} \text{ is countable.}$

Countable case 4: Set operations

Theorem. If A and B are countable, then

$$A \cup B, \quad A \cap B, \quad A \setminus B$$

are countable.



To prove the union claim, choose injections $f : A \rightarrow \mathbb{N}$ and $g : B \rightarrow \mathbb{N}$. Define

$$h : A \cup B \longrightarrow \mathbb{N} \times \{0, 1\}, \quad h(x) = \begin{cases} (f(x), 0), & x \in A, \\ (g(x), 1), & x \in B \setminus A. \end{cases}$$

The two tags prevent a collision between the two cases, so h is injective. Since $\mathbb{N} \times \{0, 1\}$ is countable, $A \cup B$ is countable.

Every subset of a countable set is countable: restrict an injection into $\mathbb{N}$ to that subset. Because

$$A \cap B \subseteq A, \quad A \setminus B \subseteq A,$$

the intersection and difference are countable as well.

Optional demo video: *How An Infinite Hotel Ran Out Of Room.*

<https://youtu.be/0xGsU8oIWjY?si=CJF3Fb0yVXWF0mlG>

4.3 Strict cardinality and uncountable sets

MUST UNDERSTAND Strict cardinality means no bijection exists; an uncountable set cannot be completely listed.

We write

$$|A| < |B|$$

when there is an injection $A \rightarrow B$ but no bijection between A and B . A standard proof has two parts:

1. construct an injection $A \rightarrow B$;
2. prove that no surjection $A \rightarrow B$ exists.

The second part rules out every bijection. Equivalently, once $|A| \leq |B|$ is known, one may prove that no injection $B \rightarrow A$ exists; otherwise Schroeder–Bernstein would give equal cardinality.

A set T is **uncountable** when it is not countable. In the usual cardinal comparison used in this course,

$$|\mathbb{N}| < |T|.$$

Informally, no sequence $t_0, t_1, t_2, \dots$ can count all elements of T one by one.

Uncountable case 1: The real numbers

We already proved $|(0, 1)| = |\mathbb{R}|$. Therefore it is enough to prove that $(0, 1)$ is uncountable. We show that every proposed list misses a number.

Suppose, for contradiction, that all numbers in $(0, 1)$ can be listed:

$$\begin{aligned} r_1 &= 0.d_{11}d_{12}d_{13}\dots \\ r_2 &= 0.d_{21}d_{22}d_{23}\dots \\ r_3 &= 0.d_{31}d_{32}d_{33}\dots \\ &\vdots \end{aligned}$$

Choose for each real its decimal expansion that does not end in an infinite string of 9s. Now construct

$$r = 0.c_1c_2c_3\dots, \quad c_i = \begin{cases} 4, & d_{ii} \neq 4, \\ 5, & d_{ii} = 4. \end{cases}$$

The number r lies in $(0, 1)$ and has no ambiguous tail of 9s. It differs from r_i in decimal position i , so $r \neq r_i$ for every i . Thus the proposed list was incomplete, a contradiction. Therefore $(0, 1)$, and hence $\mathbb{R}$, is uncountable.

The reason is not that the real line is dense. The rational numbers $\mathbb{Q}$ are also dense, yet $\mathbb{Q}$ is countable. The decisive point is the diagonal construction: it defeats every proposed list.

The proof is **Cantor diagonalization**. Its reusable pattern is:

1. Assume a complete list exists.

2. Use the i th listed object to choose the i th part of a new object.
3. Ensure the new object differs from item i at position i .
4. Conclude that the list was not complete.

Uncountable case 2: Cantor's theorem

Cantor's theorem. For every set S ,

$$|S| < |\mathcal{P}(S)|.$$

That is, the power set of S is “bigger” than S in cardinality.

The injection $x \mapsto \{x\}$ gives $|S| \leq |\mathcal{P}(S)|$. To prove the strict inequality, it remains to show that no function $f : S \rightarrow \mathcal{P}(S)$ can reach every subset in $\mathcal{P}(S)$; equivalently, no surjective f exists.

First fix any function $f : S \rightarrow \mathcal{P}(S)$. The function assigns each element $x \in S$ to one subset $f(x) \subseteq S$. Define

$$D = \{x \in S \mid x \notin f(x)\}.$$

The set D collects exactly those elements that do not belong to their own assigned subsets. It is well-defined for every f before any assumption about surjectivity, and $D \subseteq S$ gives $D \in \mathcal{P}(S)$.

For example, if $S = \{a, b, c\}$, one possible function is

$$f(a) = \{a, b\}, \quad f(b) = \{a, c\}, \quad f(c) = \{b\}.$$

For each input, ask whether that input belongs to its own assigned subset:

x	$f(x)$	$x \in f(x)?$	Put x in $D?$
a	$\{a, b\}$	yes	no
b	$\{a, c\}$	no	yes
c	$\{b\}$	no	yes

Here the rule produces $D = \{b, c\}$, which is not equal to $f(a)$, $f(b)$, or $f(c)$.

Now suppose, for contradiction, that f is surjective. Because $D \subseteq S$, we have $D \in \mathcal{P}(S)$, so surjectivity gives some $y \in S$ with $f(y) = D$. At this one input,

$$y \in D \iff y \notin f(y) \iff y \notin D,$$

a contradiction. If y belongs to D , the definition says it does not; if y does not belong to D , the definition says it does. Hence f cannot be surjective, so no bijection $S \rightarrow \mathcal{P}(S)$ exists.

There is no circular reasoning. The diagonal set is constructed from an arbitrary f . Surjectivity is used only afterward, to claim that some input y must map to that already-defined set D .

This theorem shows that there is no largest infinity: taking a power set always produces a strictly larger cardinality.

Uncountable case 3: Irrational numbers

The irrational numbers are $\mathbb{R} \setminus \mathbb{Q}$. They are uncountable. Otherwise both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ would be countable, so their union

$$\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$$

would be countable. This contradicts the uncountability of $\mathbb{R}$.

4.4 Reals and power sets

Theorem. The power set of the natural numbers has the same cardinality as the real numbers:

$$|\mathcal{P}(\mathbb{N})| = |\mathbb{R}|.$$

Because $|(0, 1)| = |\mathbb{R}|$, it is enough to compare $\mathcal{P}(\mathbb{N})$ with $(0, 1)$.

A subset $A \subseteq \mathbb{N}$ determines one bit for each natural number:

$$b_n = \begin{cases} 1, & n \in A, \\ 0, & n \notin A. \end{cases}$$

This is the connection between a power set and an infinite digit string.

For a rigorous comparison, first choose the canonical binary expansion of each $x \in (0, 1)$: use the expansion that does not end in infinitely many 1s. The positions containing 1 form a subset of $\mathbb{N}$, giving an injection $(0, 1) \rightarrow \mathcal{P}(\mathbb{N})$.

In the other direction, encode the membership bits as ternary digits:

$$A \mapsto \frac{1}{3} + \sum_{n=0}^{\infty} \frac{2b_n}{3^{n+2}} \in (0, 1).$$

Different subsets give different strings of digits 0 and 2, so this map is injective. The Schroeder–Bernstein theorem now gives $|\mathcal{P}(\mathbb{N})| = |(0, 1)| = |\mathbb{R}|$.

4.5 Infinite cardinal notation

Every countably infinite set S has the same cardinality as $\mathbb{N}$:

$$|S| = |\mathbb{N}| = \aleph_0.$$

The symbol $\aleph_0$ (“aleph null”) denotes the smallest infinite cardinality. A finite set is countable, but its cardinality is not $\aleph_0$.

The symbol $\aleph_1$ (“aleph one”) denotes the smallest uncountable cardinality:

$$\aleph_0 < \aleph_1.$$

4.6 The continuum hypothesis

Define the cardinality of the real numbers as the **continuum**:

$$\mathfrak{c} := |\mathbb{R}|.$$

We already know that $\mathfrak{c} = |\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$. Cantor's theorem gives

$$\mathfrak{c} = |\mathbb{R}| = |\mathcal{P}(\mathbb{N})| > |\mathbb{N}| = \aleph_0.$$

The **continuum hypothesis** (CH) asks a simple yet profound question:

Is the cardinality of the real numbers the **smallest uncountable cardinality**?

Equivalently,

$$\mathfrak{c} = \aleph_1 \text{ ?}$$

or: is there no cardinality strictly between the sizes of $\mathbb{N}$ and $\mathbb{R}$?

4.7 Independence from ZFC

This is optional. **ZFC** is Zermelo–Fraenkel set theory plus the Axiom of Choice, the standard axiom system for most mathematics. It is **consistent** if its axioms do not produce a contradiction.

A statement is **independent of ZFC** if ZFC proves neither it nor its negation. Assuming ZFC is consistent, both

$$\text{ZFC} + \text{CH} \quad \text{and} \quad \text{ZFC} + \neg\text{CH}$$

are consistent. Gödel established the first direction in 1940; Cohen established the second in 1963. ZFC alone therefore cannot decide CH.

5 Review exercises

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x^2 + 1$. Determine its range and decide whether it is injective or surjective. Then choose a domain and codomain that make the same rule bijective.
2. How many functions are there from a 4-element set to a 3-element set? How many are injective?
3. The even natural numbers form a proper subset of $\mathbb{N}$. Must they have smaller cardinality?
4. Explain in one sentence why the diagonal real r cannot equal the i th number in the proposed list.

Short answers

1. The range is $[1, \infty)$; the stated function is neither injective nor surjective. Restrict to $[0, \infty) \rightarrow [1, \infty)$ for a bijection.

2. $3^4 = 81$ functions and no injections.
3. No. The bijection $n \mapsto 2n$ shows that the even natural numbers and $\mathbb{N}$ have the same cardinality.
4. Its i th decimal digit was chosen to differ from the i th digit of r_i .

6 Summary

- Sets are determined by membership; their operations follow the corresponding logical operations.
- A function assigns exactly one output to each input. Injection, surjection, and bijection describe how inputs meet targets.
- Functions compare cardinalities. For infinite sets, a proper subset may have the same size as the full set.
- Countable sets can be listed. Diagonalization proves that the real numbers cannot be listed.
- Cantor's theorem gives $|S| < |\mathcal{P}(S)|$; the continuum hypothesis asks whether $\mathfrak{c} = \aleph_1$.

The next topic is basic counting: systematic methods for counting possible outcomes.